

University of Windsor

Odette School of Business

The Impact of Extreme Weather on Electricity Demand in Ontario

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October 23, 2026

Word Count: 1649

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Introduction

Climate change refers to long-term shifts in temperatures and weather patterns (United Nations, n.d.). Due to climate change, extreme weather events are occurring more often in Ontario. Extreme heat and cold might change electricity demand, especially when the demand for cooling and heating increases. If electricity suppliers cannot identify potential high-demand periods under extreme weather conditions, they might face challenges with power supply capability, resource distribution, and power grid stability. Therefore, it is essential to help electricity suppliers predict high-demand periods and have insight into the relationship between electricity demand, temperature changes, and weather-related conditions.

Aim of the project

To analyze and learn about the impact of extreme weather conditions and temperature changes on electricity demand in Ontario; develop a prediction model for identifying unusually high-demand periods; and deliver a data visualization dashboard and system showing the relationship between electricity demand and temperature changes in different locations.

Objectives of the project:

- To complete data collection, data integration, data cleaning, and data transformation based on historical electricity demand and weather data.
- To estimate the relationship between electricity consumption and temperature and identify the changing trends of this relationship.
- To analyze whether season, date, and region will have further impact on the relationship between electricity and temperature changes.
- To help stakeholders predict and identify periods of high electricity demand and become more aware of the importance of sustainability and energy conservation.
- To develop an interactive dashboard and system showing our analysis and providing references for electricity demand prediction and resource planning.

Project boundaries and scope

In scope:

- To select six different cities across Ontario to analyze and build a model to predict other areas in Ontario (Windsor-Southwestern Ontario, Toronto-Central Ontario, Ottawa-Eastern Ontario, London-Southwestern/Central Ontario, Sudbury-Northeastern Ontario, and Thunder Bay-Northwestern Ontario).
- To collect historical data on electricity demand from IESO and historical data on weather conditions from Environment and Climate Change Canada (2019-2025).
- To perform data wrangling using R and RStudio and conduct in-depth data analysis based on historical datasets.
- To store the cleaned and integrated datasets in MySQL.
- To use temperature as the primary research variable in the data analysis and analyze its potential relationship with electricity demand.
- To evaluate whether the electricity demand-weather relationship will be influenced by other available variables, such as season, date, and region.
- To develop a prediction model and establish an interactive dashboard through Tableau based on historical datasets.

Out of scope:

- To develop a real-time electricity control and monitoring system
- To advise electricity suppliers on how much electricity output they need to increase or decrease under extreme weather conditions.
- To consider every factor (such as price, population growth, and changes in consumption habits) that might affect electricity demand.
- To determine any power infrastructure investment decisions.
- To guarantee that predictions will be completely accurate under all weather conditions.

Stakeholder roles

No	Role	User Story
1	Electricity suppliers and grid operators (responsible for interpreting analytical findings and predictions to support future resource planning and allocation)	As electricity suppliers and grid operators, we want to use the analysis results by selecting season, date, and region to compare and understand the impact of extreme weather on electricity demand under different criteria, so that we can identify potential high-demand periods and prepare electricity allocation plans in advance.
2	The project team (responsible for completing data wrangling, managing the database, developing a prediction model and an interactive dashboard, and maintaining the system)	As the project team, we want to use credible, cleaned, and integrated data to conduct in-depth data analysis, so that we can ensure the dashboard and system clearly display results on the impact of extreme weather on electricity demand in Ontario.
3	Government & policymakers (responsible for preparing long-term resource planning and setting up climate change adaptation strategies based on the project findings)	As government and policymakers, we want to understand the weather-related electricity demand patterns and the impact of extreme weather conditions on electricity demand, so that we can set up long-term climate change adaptation strategies and support resource planning.
4	Consumers (responsible for enhancing environmental and energy conservation awareness)	As consumers, we want to have insight into the impact of extreme weather on electricity demand, so that we can be aware of how to protect our environment and have a more sustainable lifestyle.

Functional Requirements

Functional requirements define the operations and capabilities needed to analyze electricity demand during extreme weather conditions in Ontario. They are prioritized according to their importance to the project.

No.	Functional Requirement	Priority
1	Data Collection: The system shall import historical hourly electricity demand records from IESO and weather observations from Environment and Climate Change Canada.	High
2	Data Preparation: The system shall identify missing values, remove duplicate records, standardize data formats, and calculate daily peak electricity demand.	High
3	Data Integration and Storage: The system shall combine electricity demand and weather records using matching dates and relevant geographical information and store the processed data in a relational database.	High
4	Extreme Weather Analysis: The system shall identify extreme temperature periods using predefined thresholds and compare electricity demand during extreme and normal weather conditions.	High
5	Predictive Modelling: The system shall develop a statistical model to identify unusually high-demand periods and evaluate its performance against a calendar-based baseline.	High
6	Data Visualization: The system shall provide an interactive dashboard displaying electricity demand, temperature trends, extreme weather periods, and predictive modelling results.	High
7	Additional Weather Analysis: The system shall support the analysis of additional weather variables, such as humidity and wind speed, where reliable data is available.	Medium
8	Weather Event Integration: The system shall allow documented extreme weather events from external sources to be incorporated into the analysis.	Low

Non-Functional Requirements

Non-functional requirements establish the quality standards needed for accurate, reliable, secure, and accessible analysis of electricity demand and weather data.

No.	Non-Functional Requirement
1	Data Integrity: The system shall maintain accurate and consistent electricity demand and weather records throughout data preparation, integration, and storage.
2	Reliability: The system shall identify and document incomplete or inconsistent records and prevent invalid data from affecting analytical results.
3	Performance: The system shall process historical datasets efficiently and support interactive dashboard operations using the team's available computing resources.
4	Usability: The dashboard shall provide clearly labelled charts, appropriate measurement units, and understandable filters for users with different technical backgrounds.
5	Security and Compliance: The system shall protect database credentials, prevent unauthorized data modifications, and follow applicable data-source licensing conditions.
6	Maintainability and Reproducibility: The system shall maintain documented data sources, processing procedures, and version-controlled scripts to support future modifications and repeatable analysis.

Acceptance Criteria

The following criteria define the conditions that must be met to confirm that the system meets the project's scope, functional and non-functional requirements.

1. **Data Import:** Credible historical electricity demand and weather datasets covering 2019–2025 shall be successfully imported, with the imported records and date ranges verified against the original sources.
2. **Data Cleaning and Data Validation:** Duplicate records shall be removed, missing values shall be identified and handled according to documented rules, and data formats shall be consistent and standardized before integration.
3. **Data Integration:** Cleaned and standardized electricity demand and weather observations shall be matched using corresponding dates, with the geographical relationship between provincial demand and weather station locations representing different areas clearly documented.
4. **Daily Data Aggregation:** Daily weather summaries and daily peak electricity demand shall be calculated correctly and verified using a sample of the original records.
5. **Extreme Weather Identification:** Predefined temperature thresholds shall be documented and correctly applied to classify extreme and normal weather observations.
6. **Database Storage:** Cleaned and integrated datasets shall be successfully stored in MySQL, and sample records shall be retrieved correctly through database queries.
7. **Data Visualization:** The dashboard should be developed in Tableau. It shall display electricity demand and temperature trends using appropriate charts, accurate dates, clearly labelled axes, and measurement units.
8. **Dashboard Interaction:** Users shall be able to filter and explore the interactive dashboard by season, date, and region, where supported by the available data, with visualizations updating correctly.
9. **Predictive Model Evaluation:** The model shall generate predictions for unusually high-demand periods and report its performance on a held-out test dataset compared with a calendar-based baseline.
10. **Documentation:** Data collection, cleaning, transformation, integration, predictive modelling, dashboard development procedures, and user guide shall be documented sufficiently to allow the project team to reproduce the analysis.
11. **Meet Stakeholder Requirements:** The interactive dashboard and system shall meet our stakeholder needs, including understanding the impact of extreme weather on electricity demand, helping to identify unusual periods of high electricity demand, and setting up energy strategies and resource allocation plans based on the project findings.

Revision History

Version	Date	Author(s)	Description of Changes	Approval Status
1.0	Sep. 27, 2026	Project Team	Initial project proposal prepared and submitted.	Submitted to Instructor
1.1	Oct. 2, 2026	Project Team	Revised project proposal with additional data wrangling information and resubmitted.	Submitted to Instructor
1.2	Oct. 20, 2026	Project Team	Prepared the initial User Requirements and Analysis Document, including project scope, stakeholder roles and preliminary system requirements.	Initial Draft
1.3	Oct. 23, 2026	Project Team	Finalized project scope, stakeholder roles, prioritized functional requirements, non-functional requirements and acceptance criteria.	Pending approval from instructor

References

- Environment and Climate Change Canada. (n.d.). *Historical climate data. Government of Canada.*
    - https://climate.weather.gc.ca
- Independent Electricity System Operator. (n.d.). *Hourly demand report.*
    - https://reports-public.ieso.ca/public/Demand/
- United Nations. (n.d.). *What is climate change?*
    - https://www.un.org/climatechange/what-is-climate-change/